POLE PARITY AND DIOPHANTINE DEFINABILITY OF THE INTEGERS

Abstract. We compare two existence assertions over the rational numbers: a positive existential predicate that accepts every integer and excludes odd-order poles outside one fixed finite set of primes, and a Diophantine definition of the ordinary integers. Under three explicit arithmetic assumptions—a cyclic subgroup of a specified Mordell–Weil group, a five-point height estimate, and compatible prime-pattern representations—we prove that these assertions are equivalent. The substantive direction first constructs multiplication on elliptic point indices. Local comparisons turn a nonzero multiplication discrepancy into a height bound, which a fixed multiplication map contradicts by quadratic canonical-height growth. A support condition and a congruence of variable precision then identify each point index with its rational scalar. The converse uses an integer-defining polynomial as the pole-parity predicate and needs none of the three arithmetic inputs. All auxiliary formulas are finite and existential. The three inputs are assumptions of this paper; the equivalence alone establishes neither of the two assertions independently.

1. The question and the equivalence

A set $A \subseteq \mathbb{Q}^r$ is *Diophantine over* $\mathbb{Q}$ if a polynomial equation with rational coefficients defines membership in A after existential quantification over finitely many additional rational variables. We consider the ordinary subset $\mathbb{Z} \subseteq \mathbb{Q}$ and a weaker condition on prime valuations of accepted rational numbers. For a rational prime q, normalize $v_q(q) = 1$ and put $v_q(0) = +\infty$.

A formula is *positive existential* if it is built from polynomial equations by finite conjunctions, finite disjunctions, and existential quantifiers. Every scalar variable in such a formula ranges over $\mathbb{Q}$. We compare the following two assertions.

Assertion P: parity of poles. There exist one positive existential formula $\Phi(t)$ over $\mathbb{Q}$ and one finite set S_Φ of rational primes such that

$$\begin{gathered} \forall n \in \mathbb{Z} \quad \mathbb{Q} \models \Phi(n), \\ \forall t \in \mathbb{Q}\ \forall \text{ primes } q \notin S_\Phi : \quad (\mathbb{Q} \models \Phi(t) \text{ and } v_q(t) < 0) \Longrightarrow v_q(t) \in 2\mathbb{Z}. \end{gathered} \tag{1.1}$$

The formula and the finite set are fixed before the universal quantifiers. Thus Φ excludes odd-order poles outside S_Φ; it is not required to define integrality. We retain the name Input P when this assertion is used as the antecedent of the forward implication.

Assertion $T_\mathbb{Z}$: the scalar integers are Diophantine. There exist an integer $N \geq 1$ and a polynomial $D \in \mathbb{Z}[X, Y_1, \ldots, Y_N]$ such that

$$\forall x \in \mathbb{Q} : \qquad x \in \mathbb{Z} \quad \Longleftrightarrow \quad \exists \mathbf{y} \in \mathbb{Q}^N \ D(x, \mathbf{y}) = 0. \tag{1.2}$$

This is an assertion about the usual rational scalar x. A related construction represents an integer n by a point nP on an elliptic curve and defines the ring operations on these representations. Such a *Diophantine model* does not by itself identify a rational scalar with a point index; see [6, Definition 1.3].

The other arithmetic data are stated in Section 2. Input E specifies a cyclic subgroup $G = \mathbb{Z}P$ of the Mordell–Weil group of a fixed elliptic curve over $\mathbb{Q}(\sqrt{2})$. Input H bounds rational heights by odd contacts with five marked points. Input R supplies prime-pattern representations for the same curve and marked points. Their compatibility is part of their formulation.

Theorem 1.1 (The equivalence)**.** *Assume Inputs* E*,* H*, and* R *of Section 2, with the shared fixed data specified there. Then*

$$\mathrm{P} \quad \Longleftrightarrow \quad T_\mathbb{Z}.$$

The implication $T_\mathbb{Z} \Rightarrow$ P holds without any of the three arithmetic inputs.

Date: 4 November 2026.

The equivalence concerns the two global existence assertions above. It does not assert that a given candidate for Φ defines $\mathbb{Z}$, or that either assertion follows from E, H, and R alone. It also places no restriction on the form of a possible independent proof of $T_{\mathbb{Z}}$.

The content of the theorem is the construction from P to $T_{\mathbb{Z}}$. Fix any m_0, P satisfying Input E. The first stage constructs a Diophantine relation Mul_G with

$$\mathrm{Mul}_G(n_1P, n_2P, n_3P) \quad \Longleftrightarrow \quad n_1n_2 = n_3 \qquad (n_1, n_2, n_3 \in \mathbb{Z}).$$

Its auxiliary rational variables satisfy $\xi_1\xi_2 = \xi_3$, whereas $T_i = n_iP$ are the prescribed points. Local conditions compare ξ_i with n_i modulo selected primes. If $\delta = n_1n_2 - n_3$ is nonzero, these comparisons make every relevant odd-contact prime divide δ to a fixed power. Input H then bounds the tested coordinate heights by a constant times $\max_i |n_i|^2$. A fixed multiplication map $T \mapsto [L]T$, applied only to the points whose coordinate heights are tested, makes the canonical-height lower bound larger. The resulting contradiction excludes every incorrect multiplication tuple, including those with small indices.

The second stage constructs a Diophantine relation Eval satisfying

$$\mathrm{Eval}(x, nP) \quad \Longleftrightarrow \quad x = n \qquad (x \in \mathbb{Q},\ n \in \mathbb{Z}).$$

It combines a height bound for x in terms of the represented integer n with a congruence $v_3(x - n) \geq k$. The precision k is computed inside the already constructed integer model. When it exceeds the height bound for $x - n$, that difference must vanish. Quantifying the point code then gives (1.2). Both signs, zero indices, and the finitely many marked integer values are included.

The arithmetic formulations and the pole-parity framework originate in the dated manuscript [1]. Here E, H, and R are explicit assumptions, and their proofs are not part of the paper. Rank-one Mordell–Weil groups, formal-parameter approximations, and height growth have precedents in Diophantine definability; see, for example, [4, Lemmas 8–12]. We make no priority claim for these individual devices. The logical contribution studied here is the equivalence with the precise scalar assertion (1.2).

The prerequisites are basic algebraic number theory and the group law, local formal group, and canonical height of an elliptic curve. Appendix A proves the local estimates used in the construction; Appendix B translates point notation into rational equations; and Appendix C gives the required definition of one local ring. These are auxiliary parts of the equivalence proof, distinct from Inputs E, H, and R. The Davis–Putnam–Robinson–Matiyasevich theorem is used only after multiplication on the integer model has been established. Background on Diophantine definitions and computably enumerable sets is given in [5, Sections 4, 6, and 8].

2. The three arithmetic inputs and fixed data

All arithmetic data chosen below are fixed throughout the construction. Over $\mathbb{Q}$, positive existential definitions are equivalent to Diophantine definitions: sums of squares combine equations, products combine alternatives, and $a \neq 0$ is written $\exists b\ (ab = 1)$. We may therefore use a finite collection of named predicates, provided each has an explicit positive existential definition.

We use $\mathbb{N} = \{0, 1, 2, \ldots\}$ and the convention $v_q(0) = +\infty$. For a rational prime q, $v_q(q) = 1$. Heights are absolute logarithmic Weil heights, with natural logarithms; in particular,

$$h(u/v) = \log \max\{|u|, v\} \quad \text{if } u, v \in \mathbb{Z},\ v > 0,\ \gcd(u, v) = 1.$$

We use the familiar inequalities $h(ab) \leq h(a) + h(b)$ and $h(a + b) \leq h(a) + h(b) + \log 2$, also over number fields.

Input E: a cyclic subgroup of a Mordell–Weil group. Set

$$F = \mathbb{Q}(\sqrt{2}), \qquad d = 75 - 53\sqrt{2}, \qquad E:\ y^2 = x^3 - d^2x.$$

Write σ for the nontrivial automorphism of $F/\mathbb{Q}$. There are a fixed integer $m_0 \geq 1$ and a fixed nontorsion point $P \in E(F)$ such that

$$G := m_0E(F) = \mathbb{Z}P. \tag{2.1}$$

Here $E(F)$ is the Mordell–Weil group of rational points over F, and G is its chosen infinite cyclic subgroup of finite index. The equality concerns rational points, rather than the algebraic image of an isogeny. The *index* of $T \in G$ is the unique $n \in \mathbb{Z}$ with $T = nP$.

Input H: a height estimate from five marked points. There is a set $\mathcal{C} = \{b_0, \dots, b_4\}$ of five distinct points of $\mathbb{P}^1(\mathbb{Q})$, with $b_0 = \infty$, and fixed primitive integral linear forms

$$L_0(U, V) = V, \qquad L_i(U, V) = \alpha_i U + \beta_i V \quad (1 \le i \le 4)$$

vanishing at the corresponding points. Thus $\alpha_i \neq 0$ for $i \ge 1$. There are a finite prime set S_0 and constants $H_0, c_0 > 0$ such that

$$h(s) \le H_0 M_{S_0}(s)^{c_0} \qquad (s \in \mathbb{P}^1(\mathbb{Q}) \setminus \mathcal{C}).$$

For $s = (u : v) \in \mathbb{P}^1(\mathbb{Q}) \setminus \mathcal{C}$ and a prime q, its *normalized contact order* with b_i is

$$v_q(L_i(u, v)) - \min\{v_q(u), v_q(v)\}. \tag{2.2}$$

This is independent of the chosen homogeneous coordinates. The number $M_{S_0}(s)$ is the product of the primes $q \notin S_0$ for which at least one of these five orders is odd. With primitive integral coordinates, the orders are simply the valuations of the five nonzero form values. Thus this product records odd orders, rather than all positive contact orders.

Input R: compatible prime-pattern representations. For the curve and the same forms $L_0, \dots, L_4$ fixed in Inputs E and H, there is a finite set $\mathcal{K} \subset \mathbb{Z}_{>0}$ with the following property. For every $A \in \mathbb{Z}$ and every real lower bound $B_0 > 0$, there are integers u_1, u_2, u_3, v, with $v \neq 0$, such that

$$Av = u_1 + u_2 + u_3, \tag{2.3}$$

and each of the thirteen entries

$$v, \qquad L_i(u_j, v) \quad (1 \le i \le 4,\ 1 \le j \le 3) \tag{2.4}$$

is $\epsilon\kappa r$, with $\epsilon \in \{1, -1\}$, $\kappa \in \mathcal{K}$, and r a positive prime exceeding B_0. Every selected prime splits in F and is a prime of good reduction for both displayed curves E and E^σ. At either place above r, if d_1 and d_2 are the orders of their reduction groups, then

$$\gcd(d_1, d_2) \mid 4, \qquad r \nmid d_1 d_2. \tag{2.5}$$

The set $\mathcal{K}$ is independent of both A and the lower bound. There is no primality predicate in our formulas: primes enter the proof that integer inputs have witnesses.

The selected primes in Input R need not be distinct. Its finite catalogue may depend on the fixed curve and marked-point data, but not on A or B_0. Choose the witnesses to E and H so that R holds for those same data; all later choices use this one configuration. This is the meaning of the joint assumptions E, H, and R in Theorem 1.1. No proof of any of these three inputs is required or supplied in the argument below.

2.1. **One exceptional set and one height constant.** For the forward implication, assume P and fix one formula Φ and one finite set S_Φ as in (1.1). These choices remain fixed through Section 5. Choose an integer $c \ge \max\{1, c_0\}$. Since $M_{S_0}(s) \ge 1$, the bound in Input H remains valid with exponent c. Choose a finite prime set S containing S_0, S_Φ, 2, 3, the primes ramified in F, all bad primes for the two displayed integral equations, every prime at most $c + 1$, and the prime divisors of

$$\prod_{i=1}^{4} \alpha_i, \qquad \prod_{0 \le i < j \le 4} \det(L_i, L_j), \qquad \prod_{\kappa \in \mathcal{K}} \kappa.$$

The determinants are nonzero because the marked points are distinct. Enlarging the exceptional set discards at most one factor q for each newly excluded prime. We may therefore fix an integer $H_* \ge 1$ such that

$$h(s) \le H_* M_S(s)^c \qquad (s \in \mathbb{P}^1(\mathbb{Q}) \setminus \mathcal{C}). \tag{2.6}$$

For instance any integer at least $H_0(\prod_{q \in S \setminus S_0} q)^c$ suffices after enlarging it to be at least 1. Neither S nor H_* will vary with an integer index or a requested congruence precision. The symbol H names the arithmetic input; H_* is the numerical coefficient in (2.6).

The canonical-height comparison and the nontorsion of P give a fixed integer $a_* \geq 1$ such that

$$h(x(tP)) \geq \frac{t^2}{a_*} - a_* \qquad (t \in \mathbb{Z} \setminus \{0\}). \tag{2.7}$$

Indeed, the canonical height of tP is $t^2\widehat{h}(P)$, with $\widehat{h}(P) > 0$, and its difference from a fixed positive multiple of $h(x(tP))$ is bounded [3, Chapter VIII]; see also the height calculation in the proof of [4, Lemma 8].

2.2. **Point notation inside rational formulas.** We encode $u_0 + u_1\sqrt{2} \in F$ by $(u_0, u_1) \in \mathbb{Q}^2$. A point has an identity flag and affine coordinates, giving a unique rational code for every point of $E(F)$. Membership in G is the existential condition $T = [m_0]U$ for some $U \in E(F)$. Appendix B gives equations for these codes and the group operations. Thus a quantified point in G always means a finite rational tuple with these conditions imposed. Fixed integer multiples and conjugation are allowed in this notation. Multiplication by a *variable* integer is not allowed until Mul_G has been constructed.

On nonidentity points of G put $z = -x/y$, and set $z(O) = 0$. A point of the displayed curve with $x = 0$ or $y = 0$ is a point of order two. Since G is torsion-free, x, y, z are nonzero on $G \setminus \{O\}$. The same applies to the conjugate group. For a finite place w write F_w for the completion, $\mathcal{O}_{F_w}$ for its valuation ring, and $\mathcal{O}_{F,w} = F \cap \mathcal{O}_{F_w}$ for the corresponding localization of $\mathcal{O}_F$. At good reduction a point is *formal at* w if it lies in the kernel of reduction; the parameter z then belongs to the maximal ideal. On a nonidentity point this is equivalent to $w(1/x) > 0$.

Let ℓ_k be the degree-k truncation of the formal logarithm in the parameter z, for an integer $k \geq 1$. Its coefficients are computed from the fixed equation. Only ℓ_c, a fixed polynomial, will occur in the multiplication formula. Infinite series are used in proofs of estimates, not as conditions in an existential formula. We also use the rational projection

$$\rho : F \longrightarrow \mathbb{Q}, \qquad \rho(u_0 + u_1\sqrt{2}) = u_0 = \tfrac{1}{2}\,\mathrm{Tr}_{F/\mathbb{Q}}(u_0 + u_1\sqrt{2}).$$

All constants and all polynomials used below are fixed finite data. The results assert the existence of the resulting formulas and algorithms; they do not supply numerical values for every imported constant or an expanded coefficient list for the final polynomial.

3. Square conditions and local divisibility

Throughout this section and through Section 5, we use P as the antecedent and E, H, and R as the arithmetic assumptions.

The pole-parity predicate can control denominators without defining an integrality condition. The useful observation is that a denominator of the form $1 + rW$ is forced to be a unit at the primes where r has odd valuation. We first give a rational version and then a version selecting one place of the quadratic field. All the conditions below are existential conditions on rational variables; valuations occur only in the statements and proofs of their properties.

Lemma 3.1. *Suppose $\Phi(r)$ and $\Phi(ru^2)$ hold, and let $q \notin S_\Phi$ have odd valuation $e = v_q(r)$. Then $e > 0$ and*

$$v_q(u) \geq -\left\lfloor \frac{e}{2} \right\rfloor.$$

Proof. The condition $\Phi(r)$ excludes a negative odd value of e. If $u \neq 0$, the valuation $e + 2v_q(u)$ is odd, so $\Phi(ru^2)$ requires it to be nonnegative. This gives the bound. The case $u = 0$ uses the convention $v_q(0) = +\infty$. □

3.1. **Several rational targets with one denominator.**

Definition 3.2. For a fixed positive integer t, define $\mathrm{Div}(r; D_1, \ldots, D_t)$, with $r, D_i \in \mathbb{Q}$, by requiring $r \neq 0$, $\Phi(r)$, and rational witnesses $W, X_1, \ldots, X_t$ such that, on writing $Y = 1 + rW$,

$$\begin{aligned} &Y \neq 0, \qquad D_iY = rX_i \quad (1 \leq i \leq t), \\ &\Phi(rW^2), \qquad \Phi(rX_i^2) \quad (1 \leq i \leq t). \end{aligned} \tag{3.1}$$

The length t is fixed in each use of this definition, rather than a variable in the formula.

Lemma 3.3 (Soundness)**.** *If* $\mathrm{Div}(r; D_1, \dots, D_t)$ *holds and* $q \notin S_\Phi$ *has odd* $v_q(r)$*, then*

$$v_q(r) > 0, \qquad v_q(D_i) \geq 1 \quad (1 \leq i \leq t).$$

Proof. Put $e = v_q(r)$ and $d = \lfloor e/2 \rfloor$. Lemma 3.1 gives $e > 0$ and valuations at least $-d$ for $W, X_1, \dots, X_t$. Thus $v_q(rW) \geq e - d > 0$, so $Y = 1 + rW$ is a q-adic unit. Equation (3.1) gives $v_q(D_i) \geq e - d \geq 1$, including zero targets under the infinite-valuation convention. □

Lemma 3.4 (Completeness for integer moduli)**.** *Let* $m \in \mathbb{Z} \setminus \{0\}$ *and* $D_1, \dots, D_t \in \mathbb{Q}$ *satisfy*

$$v_p(D_i) \geq v_p(m) \qquad (p \mid m,\ 1 \leq i \leq t). \tag{3.2}$$

Then $\mathrm{Div}(m; D_1, \dots, D_t)$ *has witnesses* $W, X_1, \dots, X_t$ *in* $\mathbb{Z}$*. For* $m = \pm 1$*, no condition on the targets is needed.*

Proof. Condition (3.2) permits a common denominator

$$D_i/m = A_i/B, \qquad A_i, B \in \mathbb{Z}, \quad B \neq 0, \quad \gcd(B, m) = 1.$$

Choose a nonzero integer z with $zB \equiv 1 \pmod{m}$, and set

$$W = \frac{zB - 1}{m}, \qquad X_i = zA_i.$$

Then $Y = 1 + mW = zB \neq 0$ and $D_i Y = mX_i$. All witnesses and all arguments of Φ are integers, so Input P supplies the required conditions. This works for negative m as well. When $m = \pm 1$, take $z = 1$; the same construction applies to arbitrary rational targets. □

3.2. **A selected place of the quadratic field.** For rational r, b, C, impose the root conditions

$$r \neq 0, \qquad \Phi(r), \qquad b^2 - 2 = rC, \qquad \Phi(rC^2). \tag{3.3}$$

They will be shared by all local conditions attached to one occurrence of r. The parameter b selects a residue of $\sqrt{2}$.

Definition 3.5. For a fixed integer $k \geq 1$ and $D \in F$, let $\mathrm{Loc}_{r,b,k}(D)$ assert the existence of $X_0, X_1, V, W \in \mathbb{Q}$ such that

$$\begin{gathered} X = X_0 + X_1\sqrt{2}, \qquad Y = 1 + rW + V(\sqrt{2} - b) \neq 0, \qquad DY = r^k X, \\ \Phi(rX_0^2), \quad \Phi(rX_1^2), \quad \Phi(rV^2), \quad \Phi(rW^2). \end{gathered} \tag{3.4}$$

The root conditions (3.3) are imposed separately. An F-equation is interpreted as two rational equations in the basis $1, \sqrt{2}$, as in Appendix B.

Lemma 3.6 (Soundness at the selected place)**.** *Assume* (3.3)*, and let* $q \notin S_\Phi \cup \{2\}$ *have odd valuation* $e = v_q(r)$*. Then* q *splits in* F*. At the place* $w \mid q$ *where* $\sqrt{2}$ *reduces to* b*, normalized by* $w(q) = 1$*,*

$$\mathrm{Loc}_{r,b,k}(D) \quad \Longrightarrow \quad w(D) \geq k.$$

The same place works for every condition using these root data.

Proof. Write $d = \lfloor e/2 \rfloor$. Lemma 3.1 gives $e > 0$ and $v_q(C) \geq -d$, whence

$$v_q(b^2 - 2) \geq e - d > 0.$$

Consequently b is a q-adic unit and 2 is a square modulo q. Since $q \neq 2$, it splits in F. At the selected place, the factor $\sqrt{2} + b$ is a unit, so

$$w(\sqrt{2} - b) = v_q(b^2 - 2) \geq e - d.$$

For witnesses in (3.4), the same lemma gives valuations at least $-d$ for X_0, X_1, V, W. Therefore

$$w(rW) \geq e - d > 0, \qquad w\big(V(\sqrt{2} - b)\big) \geq e - 2d = 1.$$

The denominator Y is a unit. Since $w(X) \geq -d$, the equation $DY = r^k X$ yields

$$w(D) \geq ke - d \geq k.$$

The last inequality uses $e \geq 1$ and $k \geq 1$; zero targets cause no exception. The selected place depends only on q and the shared value of b. □

Lemma 3.7 (Completeness at a split prime)**.** *Let r be an odd rational prime split in F, let $w \mid r$, and choose $b \in \mathbb{Z}$ reducing to $\sqrt{2}$ at w. Put $C = (b^2-2)/r \in \mathbb{Z}$. For every $D \in r^k \mathcal{O}_{F,w}$, the root conditions (3.3) hold and $\mathrm{Loc}_{r,b,k}(D)$ has witnesses $X_0, X_1, V, W \in \mathbb{Z}$.*

Proof. Since $\mathcal{O}_F = \mathbb{Z}[\sqrt{2}]$, write

$$D/r^k = A/B, \qquad A, B \in \mathcal{O}_F, \quad B \notin \mathfrak{p}_w.$$

Write $B = B_0 + B_1\sqrt{2}$ and $B_b = B_0 + bB_1$. The integer B_b is nonzero modulo r. Choose $z \in \mathbb{Z}$ with $zB_b \equiv 1 \pmod{r}$; then $z \neq 0$. Take

$$X = zA, \qquad V = zB_1, \qquad W = \frac{zB_b - 1}{r}.$$

The denominator in (3.4) is exactly

$$1 + rW + V(\sqrt{2} - b) = zB_b + zB_1(\sqrt{2} - b) = zB.$$

It is nonzero, and $DY = r^k X$. The coefficients of X, as well as V, W, C, are integers. Input P therefore supplies every occurrence of Φ. □

These definitions impose square conditions only on the witnesses explicitly listed. Witnesses internal to Φ, and reciprocal witnesses used to express inequations, are not subjected to further conditions. Each definition is therefore a finite existential formula. The soundness lemmas are one-way implications; the completeness lemmas state exactly which intended instances admit witnesses.

4. Multiplication from quadratic height growth

We now construct one formula for multiplication of integer indices. Its auxiliary rational values satisfy $\xi_1\xi_2 = \xi_3$. Local comparisons will show that a nonzero discrepancy $n_1n_2 - n_3$ bounds certain coordinate heights. Applying a fixed multiplication map to the points makes the quadratic lower height bound incompatible with that discrepancy. The local precision in this entire construction is the fixed integer c. Write ℓ_c for the degree-c truncation of the formal logarithm of E, as in Appendix A.

4.1. **Local comparisons.**

For U on E or E^σ, write

$$\mathrm{For}_{r,b}(U): \quad U = O \quad \text{or} \quad \big(U \neq O,\ x(U)y(U) \neq 0,\ \mathrm{Loc}_{r,b,1}(1/x(U))\big).$$

At a prime outside S with odd $v_q(r)$, the root conditions and Lemma 3.6 make this a sufficient condition for U to reduce to O at the selected place. We use $z(O) = 0$ on the identity branch. All other quotients are accompanied by their nonzero-denominator conditions. For nonidentity points in G or G^σ, the conditions $x, y \neq 0$ are automatic: a point with either coordinate zero is two-torsion.

Definition 4.1. Let $\boldsymbol{\xi} = (\xi_1, \xi_2, \xi_3) \in \mathbb{Q}^3$ and $\mathbf{T} = (T_1, T_2, T_3) \in G^3$. Put $\xi_0 = 1$ and $T_0 = P$. The condition $\mathrm{Cmp}(r; \boldsymbol{\xi}, \mathbf{T})$ requires root data b, C satisfying (3.3), and points

$$J, R_0, R_1, R_2, R_3 \in G, \qquad J \neq O, \qquad \ell_c(z(R_0)) \neq 0.$$

Set $V_i = R_i - [4]T_i$. For every $i = 0, 1, 2, 3$, impose

$$\begin{gathered} \mathrm{For}_{r,b}(R_i), \quad \mathrm{For}_{r,b}(J^\sigma), \quad \mathrm{For}_{r,b}(V_i^\sigma), \\ \mathrm{Loc}_{r,b,c}\left(\frac{\ell_c(z(R_i))}{\ell_c(z(R_0))} - \xi_i\right), \qquad \mathrm{Loc}_{r,b,c}\left(\frac{z(V_i^\sigma)}{z(J^\sigma)}\right). \end{gathered} \tag{4.1}$$

The local conditions share r, b, C but otherwise have separate witnesses.

Lemma 4.2. *Write $T_i = n_i P$ for $i = 1, 2, 3$.*

(1) If $\mathrm{Cmp}(r; \boldsymbol{\xi}, \mathbf{T})$ *holds, then, for every prime $q \notin S$ with odd $v_q(r)$,*

$$v_q(\xi_i - n_i) \geq c \quad (i = 1, 2, 3). \tag{4.2}$$

(2) If $\xi_i = n_i \in \mathbb{Z}$ and $r \notin S$ is a prime supplied by Input R, then $\mathrm{Cmp}(r; \boldsymbol{\xi}, \mathbf{T})$ *holds.*

Proof. For the first assertion, Lemma 3.6 selects one place $w \mid q$ at which all local conditions are sound. Write $R_i = t_i P$ and $J = jP$, where $j \neq 0$, and set $n_0 = 1$. The points tested by For lie in their formal subgroups at w. The exact valuation comparison of Lemma A.3, applied on E^σ, gives

$$v_q\left(\frac{t_i - 4n_i}{j}\right) \geq c \qquad (0 \leq i \leq 3).$$

Because j is an integer, $t_i \equiv 4n_i \pmod{q^c}$. The case $i = 0$ gives $t_0 \equiv 4 \pmod{q^c}$. Since $q \neq 2$, the denominator index is a unit and

$$t_i/t_0 \in \mathbb{Z}_q, \qquad t_i/t_0 \equiv n_i \pmod{q^c}.$$

Now $q > c + 1$, by the choice of S. The truncated-logarithm estimate in Lemma A.3 and the first local quotient in (4.1) show that $v_q(\xi_i - n_i) \geq c$. In particular every ξ_i is q-integral. The argument includes zero numerator points under the convention $z(O) = 0$.

For completeness, choose either place $w \mid r$. Let d_1, d_2 be the orders of the reductions of E, E^σ there. Input R gives $\gcd(d_1, d_2) \mid 4$ and $r \nmid d_1 d_2$, so there is a positive integer h with

$$h \equiv 0 \pmod{d_1}, \qquad h \equiv 4 \pmod{d_2 r^c}.$$

Take

$$J = d_2 P, \qquad R_i = h n_i P \quad (0 \leq i \leq 3).$$

Here $n_0 = 1$, so $R_0 = hP$, and $V_i = (h - 4)n_i P$. These choices put every R_i in the formal subgroup of E and J^σ, V_i^σ in that of E^σ. The truncated-logarithm estimate gives

$$w\left(\frac{\ell_c(z(R_i))}{\ell_c(z(R_0))} - n_i\right) \geq c,$$

and the exact valuation comparison gives

$$w\left(\frac{z(V_i^\sigma)}{z(J^\sigma)}\right) = v_r\left(\frac{(h - 4)n_i}{d_2}\right) \geq c.$$

Both denominator points are nonidentity. The same formal estimate ensures $\ell_c(z(R_0)) \neq 0$. Choose one integer lift b of $\sqrt{2}$ at w, and put $C = (b^2 - 2)/r$. Lemma 3.7 supplies all conditions in (4.1), including the depth-one conditions on reciprocal x-coordinates, with this shared choice. Identity points use their identity branches. □

4.2. **Representations and the discrepancy bound.**

Definition 4.3. The condition $\mathrm{Rep}(A; \boldsymbol{\xi}, \mathbf{T})$ requires $u_1, u_2, u_3, v \in \mathbb{Q}$ with $v \neq 0$ and

$$Av = u_1 + u_2 + u_3.$$

For each of the thirteen entries

$$v, \qquad L_i(u_j, v) \quad (1 \leq i \leq 4,\ 1 \leq j \leq 3), \tag{4.3}$$

require an expression $\epsilon \kappa r$, with $\epsilon \in \{1, -1\}$, $\kappa \in \mathcal{K}$, $r \in \mathbb{Q}^\times$, and impose $\mathrm{Cmp}(r; \boldsymbol{\xi}, \mathbf{T})$. Different occurrences have separate witnesses. The finite choices of ϵ, κ are finite disjunctions.

Lemma 4.4. *For every $A, n_1, n_2, n_3 \in \mathbb{Z}$, the condition* $\mathrm{Rep}(A; \boldsymbol{\xi}, \mathbf{T})$ *holds with $\xi_i = n_i$ and $T_i = n_i P$.*

Proof. Apply Input R to A, taking every selected prime outside the fixed finite set S. Apply the completeness assertion of Lemma 4.2 to each occurrence in (4.3). □

Lemma 4.5 (A height bound from a nonzero discrepancy)**.** *Suppose* $\xi_1\xi_2 = \xi_3$*,* $T_i = n_iP$*, and* $\delta = n_1n_2 - n_3 \neq 0$*. Then*

$$\mathrm{Rep}(A; \boldsymbol{\xi}, \mathbf{T}) \quad \Longrightarrow \quad h(A) \leq 3H_*|\delta| + 2. \tag{4.4}$$

Proof. Let $s_j = (u_j : v)$ be one of the three points in a representation. All five defining linear forms are nonzero there, so $s_j \notin \mathcal{C}$. Suppose q occurs in $M_S(s_j)$, and put $\mu = \min\{v_q(u_j), v_q(v)\}$. The minimum valuation among the five linear-form values is μ: their coefficients are integral, and a pair with q-unit determinant recovers u_j, v. Since one normalized contact order is odd, the valuations of these five values cannot all have the same parity. At least one is odd. Its expression $\epsilon\kappa r$ has odd $v_q(r)$, since the prime divisors of every $\kappa \in \mathcal{K}$ lie in S.

Lemma 4.2 gives $\xi_i \in \mathbb{Z}_q$ and $v_q(\xi_i - n_i) \geq c$ for all three indices. The identity

$$\delta = (n_1 - \xi_1)n_2 + \xi_1(n_2 - \xi_2) + (\xi_3 - n_3)$$

then gives $v_q(\delta) \geq c$. Thus

$$M_S(s_j)^c \mid \delta.$$

The height estimate (2.6) implies $h(s_j) \leq H_*|\delta|$. Finally $A = u_1/v + u_2/v + u_3/v$, so

$$h(A) \leq \sum_{j=1}^{3} h(s_j) + \log 3 \leq 3H_*|\delta| + 2.$$

□

4.3. **The multiplication formula.** Fix a positive integer L such that

$$L^2 > 18a_*H_* + a_*^2 + 8a_*. \tag{4.5}$$

All constants are now fixed. Multiplication by L on the elliptic curve is a fixed algebraic operation, obtained by finitely many group additions. It does not use multiplication of variable integer indices.

Definition 4.6. For point codes T_1, T_2, T_3, require $T_i \in G$ and rational ξ_1, ξ_2, ξ_3 with $\xi_1\xi_2 = \xi_3$. For each $j = 1, 2, 3$, require either $T_j = O$, or witnesses $Q_j \in G \setminus \{O\}$ and $A_{j,0}, A_{j,1}, A_{j,2} \in \mathbb{Q}$ satisfying

$$\begin{gathered} Q_j = [L]T_j, \qquad A_{j,2} \neq 0, \qquad x(Q_j) = \frac{A_{j,0} + A_{j,1}\sqrt{2}}{A_{j,2}}, \\ \mathrm{Rep}(A_{j,t}; \boldsymbol{\xi}, \mathbf{T}) \quad (t = 0, 1, 2). \end{gathered} \tag{4.6}$$

Call the resulting condition $\mathrm{Mul}_G(T_1, T_2, T_3)$. Every representation in (4.6) compares the full original tuple $\mathbf{T}$, not the amplified points Q_j.

Membership in $G = m_0E(F)$, fixed group operations, and the coordinate conditions are existential by Appendix B. All other predicates in the definition have fixed finite length. Consequently Mul_G is one existential formula over $\mathbb{Q}$.

Theorem 4.7 (Multiplication on the cyclic group)**.** *For all* $n_1, n_2, n_3 \in \mathbb{Z}$*,*

$$\mathbb{Q} \models \mathrm{Mul}_G(n_1P, n_2P, n_3P) \quad \Longleftrightarrow \quad n_1n_2 = n_3. \tag{4.7}$$

Together with the elliptic group law, this gives a Diophantine model of the ring $\mathbb{Z}$ *on the rational point codes of the original group* G*, under* $n \mapsto nP$*.*

Proof. If $n_1n_2 = n_3$, take $\xi_i = n_i$. For every nonidentity $T_j = n_jP$, the point $Q_j = [L]T_j$ is nonidentity, since G is torsion-free. Its x-coordinate has a representation in (4.6) with all three $A_{j,t}$ integral. Lemma 4.4 supplies the required witnesses. For $T_j = O$, use the identity branch. This proves completeness for positive, negative, and zero indices.

Conversely, suppose the formula holds with $T_i = n_iP$ and set $\delta = n_1n_2 - n_3$. Assume $\delta \neq 0$. Then

$$B = \max\{|n_1|, |n_2|, |n_3|\} \geq 1, \qquad |\delta| \leq B^2 + B \leq 2B^2.$$

Choose j with $|n_j| = B$. Its branch supplies $Q_j = Ln_jP$ and three coordinate representatives. Lemma 4.5 and the height inequalities over F give

$$\begin{aligned} h(x(Q_j)) &\le \sum_{t=0}^{2} h(A_{j,t}) + h(\sqrt{2}) + \log 2 \\ &\le 9H_*|\delta| + 8 \le 18H_*B^2 + 8. \end{aligned} \tag{4.8}$$

Here $h(\sqrt{2}) + \log 2 = \frac{3}{2}\log 2 < 2$. On the other hand, (2.7) gives

$$h(x(Q_j)) \ge \frac{L^2B^2}{a_*} - a_*.$$

Combining these inequalities yields

$$\left(\frac{L^2}{a_*} - 18H_*\right) B^2 \le a_* + 8,$$

contrary to (4.5) and $B \ge 1$. Thus $\delta = 0$, proving (4.7).

The set of point codes for G and its addition law are existentially defined; O represents zero and P represents one. The computable bijection $n \mapsto nP$ therefore transports both ring operations to Diophantine relations on G. □

Remark 4.8. The auxiliary rational values need not be the integer indices. For $T_1 = T_2 = T_3 = O$, every coordinate condition takes its identity branch, and

$$(\xi_1, \xi_2, \xi_3) = (1/2, 1/2, 1/4)$$

satisfies the remaining equation $\xi_1\xi_2 = \xi_3$. These nonintegral witnesses represent the true index relation $0 \cdot 0 = 0$; they do not evaluate the indices as rational scalars.

4.4. **Polynomial and computably enumerable relations.**

Theorem 4.9. *For every $f \in \mathbb{Z}[X_1, \dots, X_r]$, one can effectively construct an existential formula $\Theta_f(T_1, \dots, T_r)$ over $\mathbb{Q}$ such that*

$$\mathbb{Q} \models \Theta_f(n_1P, \dots, n_rP) \quad \Longleftrightarrow \quad f(n_1, \dots, n_r) = 0.$$

The formula requires all its free point arguments to belong to G.

Proof. Choose an arithmetic circuit for f. Introduce a point in G for each gate. An integer constant d is represented by the fixed point $[d]P$; an addition gate uses the group law; a multiplication gate uses Mul_G. Require the output to be O. Induction through the circuit shows that the index at each gate is the value of its corresponding integer expression. Thus the output condition is equivalent to $f(\mathbf{n}) = 0$ for the prescribed input indices. Conversely, those integer gate values supply point witnesses whenever $f(\mathbf{n}) = 0$.

The point $[d]P$ is computable from d, and each gate uses a fixed existential relation. This makes the construction effective once the fixed arithmetic data have been chosen. Constant polynomials are covered by constant output gates, while the input point conditions remain in force. □

Corollary 4.10. *For every computably enumerable set $A \subseteq \mathbb{Z}^r$, the relation*

$$\widehat{A} = \{(n_1P, \dots, n_rP) : (n_1, \dots, n_r) \in A\}$$

is Diophantine over $\mathbb{Q}$.

Proof. The Davis–Putnam–Robinson–Matiyasevich theorem [2, Theorem 8.1] gives a polynomial representation over the integers. To spell out the signed-integer convention, apply its nonnegative-integer version to the computably enumerable set

$$\{(\mathbf{u}, \mathbf{v}) \in \mathbb{N}^r \times \mathbb{N}^r : \mathbf{u} - \mathbf{v} \in A\}.$$

The positive and nonnegative versions differ only by a shift. Impose $n_i = u_i - v_i$ and express each nonnegative variable by a sum of four squares of unrestricted integers. Combining the resulting finite system by a sum of squares gives an integer polynomial $U(\mathbf{n}, \mathbf{w})$ with

$$\mathbf{n} \in A \quad \Longleftrightarrow \quad \exists \mathbf{w} \in \mathbb{Z}^s \; U(\mathbf{n}, \mathbf{w}) = 0.$$

Use Theorem 4.9 for U, retain the point arguments representing $\mathbf{n}$, and existentially quantify the point arguments representing $\mathbf{w}$. □

The arithmetic inputs and the multiplication predicate are fixed throughout. The formula for a particular f or A may have a different number of gate or witness variables; no variable-length list occurs inside any one formula.

5. Recovering the integer represented by a point

Theorem 4.7 supplies arithmetic on the points nP. We now recover the corresponding scalar $n \in \mathbb{Q}$. The argument combines a height bound with a congruence: if $x \neq n$ and $v_3(x - n) \geq k$, then

$$k \log 3 \leq h(x - n) \leq h(x) + h(n) + \log 2.$$

Thus a bound for $h(x)$ in terms of the represented integer allows a sufficiently precise congruence to force equality. We construct both conditions by fixed existential formulas. The exceptional set S and the positive integers H_*, c are those fixed in Section 2.

Throughout this section use the rational coefficient projection

$$\rho : F \longrightarrow \mathbb{Q}, \qquad \rho(a_0 + a_1\sqrt{2}) = a_0 = \frac{1}{2}\operatorname{Tr}_{F/\mathbb{Q}}(a_0 + a_1\sqrt{2}).$$

The same notation denotes its extension to $F \otimes_{\mathbb{Q}} \mathbb{Q}_p = \prod_{w|p} F_w$. This is a continuous $\mathbb{Q}_p$-linear map. As before, $z(O) = 0$.

5.1. **A support condition.** A nonidentity point of G has nonzero x- and y-coordinates: on the displayed equation a point with either coordinate zero has order two, whereas G is torsion-free. The rational functions below are therefore defined on their stated domains.

Definition 5.1. For $r \in \mathbb{Q}$ and a point code T, let $\mathrm{Supp}(r, T)$ assert that $r \neq 0$, $T \in G \setminus \{O\}$, and there exist $J, R \in G \setminus \{O\}$ and $a, b, D \in \mathbb{Q}$ such that

$$\mathrm{Mul}_G(T, J, R), \qquad \frac{1}{x(J)} = a + b\sqrt{2}, \qquad D = \rho\left(\frac{z(R)}{z(J)}\right), \qquad \mathrm{Div}(r; a, b, D). \tag{5.1}$$

All rational functions are encoded together with their nonzero denominators.

This is one finite existential formula. If $T = mP$ and $J = jP$, its multiplication condition says $R = mjP = [m]J$.

Proposition 5.2. *The support relation has the following two properties.*

(1) If $m \in \mathbb{Z} \setminus \{0\}$ and $\mathrm{Supp}(r, mP)$ holds, then, for every prime $q \notin S$,

$$v_q(r) \text{ odd} \quad \Longrightarrow \quad v_q(r) > 0 \text{ and } q \mid m. \tag{5.2}$$

(2) For every $m \in \mathbb{Z} \setminus \{0\}$,

$$\mathrm{Supp}(m, mP). \tag{5.3}$$

Proof. For the first assertion, fix $q \notin S$ with $v_q(r)$ odd. Lemma 3.3 gives

$$v_q(r) > 0, \qquad v_q(a), v_q(b), v_q(D) \geq 1.$$

At every place $w \mid q$, normalized by $w(q) = 1$, the two coefficient bounds imply $w(1/x(J)) \geq 1$. The curve has good reduction there, so J lies in the formal subgroup. Its multiple $R = [m]J$ lies in that subgroup as well. By Lemma A.1,

$$[m](Z) = mZ + Z^2 F_m(Z), \qquad F_m(Z) \in \mathcal{O}_{F_w}[[Z]]$$

for every integer m, including negative integers. Since $w(z(J)) \geq 1$, it follows that

$$w\left(\frac{z(R)}{z(J)} - m\right) \geq 1.$$

These inequalities hold at every place above q. Because $q \neq 2$, applying ρ gives $v_q(D - m) \geq 1$. Together with $v_q(D) \geq 1$, this implies $q \mid m$.

For the second assertion, first suppose $|m| > 1$. At each place above each prime $p \mid m$, choose a sufficiently small open formal subgroup about O. For the fixed integer m, the quotient

$$u_m(J) = \frac{z([m]J)}{z(J)}$$

extends analytically to O with value m, by the same multiplication series. Also $1/x(J)$ tends to zero as J tends to O. The two rational coefficient projections and ρ are continuous on $F \otimes_{\mathbb{Q}} \mathbb{Q}_p$. Consequently the subgroups can be chosen so that, whenever J lies in all the chosen neighborhoods above p,

$$v_p(a), v_p(b), v_p(\rho(u_m(J)) - m) \geq v_p(m).$$

These are simultaneous conditions at all places above p. They apply also at bad and ramified primes: the integral formal group and its multiplication series are available sufficiently close to O, as recorded in Lemma A.1. Fixed denominators in the coefficient projections merely require smaller neighborhoods.

Each chosen open subgroup has finite index in the compact local group $E(F_w)$. A common nonzero integer multiple $J = jP$ therefore belongs to all of them. Both J and $R = [m]J$ are nonidentity, and for the resulting rational quantities we have

$$v_p(a), v_p(b), v_p(D) \geq v_p(m) \qquad (p \mid m).$$

Lemma 3.4, applied to the triple (a, b, D) with modulus m, supplies the required witnesses. The model multiplication condition holds because $R = [m]J$.

For $m = \pm 1$, choose any nonidentity $J \in G$ and put $R = [m]J$. The local hypotheses of Lemma 3.4 are empty for these moduli, so it again supplies the triple certificate. □

The support condition concerns odd valuations outside S. It is not an equality test: $\mathrm{Supp}(1, mP)$ holds for every nonzero integer m, by the same argument with modulus 1 and any nonidentity J. Only the two properties in Proposition 5.2 will be used.

5.2. **Congruences with variable precision.** We use the fixed existential definition of $\mathbb{Z}_{(3)} = \{u \in \mathbb{Q} : v_3(u) \geq 0\}$ supplied by Lemma C.1. To produce a congruence of variable precision, we first choose one fixed formal point.

The prime 3 is inert and unramified in F. Moreover,

$$\mathrm{N}_{F/\mathbb{Q}}(d) = 75^2 - 2 \cdot 53^2 = 7, \qquad \Delta(E) = 64d^6,$$

so the displayed equation has good reduction at the unique place $w \mid 3$. Write $\mathbb{F}_9 = \mathbb{F}_3(\alpha)$ with $\alpha^2 = 2$. Since $d \equiv \alpha$, the reduced equation is

$$y^2 = x^3 + x.$$

The map $x \mapsto x^3 + x$ is the nonzero $\mathbb{F}_3$-linear trace $\mathbb{F}_9 \to \mathbb{F}_3$, with three preimages of each value. The values $0, 1, 2$ give respectively $3, 6, 6$ affine points: each nonzero element of $\mathbb{F}_3$ is a square in $\mathbb{F}_9$, since $1 = 1^2$ and $2 = \alpha^2$. Including the point at infinity gives

$$\#\widetilde{E}(\mathbb{F}_9) = 3 + 6 + 6 + 1 = 16.$$

Thus the fixed nonidentity point $U = [16]P$ lies in the formal subgroup at w. Normalize $w(3) = 1$.

For $n \in \mathbb{Z}$ and $k \in \mathbb{N}$, put

$$B_k = [3^k]U = [16 \cdot 3^k]P, \qquad A_{n,k} = [n]B_k, \tag{5.4}$$

and define rational numbers

$$c_{n,k} = \rho\left(\frac{z(A_{n,k})}{z(B_k)}\right), \qquad b_k = \rho\left(\frac{z(B_k)}{z(U)}\right). \tag{5.5}$$

The points B_k are nonidentity. The convention $z(O) = 0$ applies when $n = 0$.

Lemma 5.3. *For every $n \in \mathbb{Z}$ and $k \in \mathbb{N}$,*

$$v_3(c_{n,k} - n) \geq k + 1, \qquad v_3(b_k) = k. \tag{5.6}$$

In particular $b_k \neq 0$ and $b_0 = 1$.

Proof. Let $\mathcal{O}_w$ be the valuation ring of F_w, and let ℓ be the formal logarithm. Lemma A.2 gives $\ell(Z)/Z \in 1 + 3\mathcal{O}_w$ for every nonzero formal parameter Z at w. The logarithm identity $\ell(z(B_k)) = 3^k \ell(z(U))$ therefore gives

$$\frac{z(B_k)}{z(U)} = 3^k \frac{\ell(z(U))/z(U)}{\ell(z(B_k))/z(B_k)} \in 3^k(1 + 3\mathcal{O}_w).$$

The projection ρ maps $1 + 3\mathcal{O}_w$ into $1 + 3\mathbb{Z}_3$, since it is the trace divided by the 3-adic unit 2. This proves $v_3(b_k) = k$. The case $k = 0$ also gives $b_0 = \rho(1) = 1$. The same identity shows

$$w(z(B_k)) = w(z(U)) + k \geq k + 1.$$

For any integer n, the integral multiplication series of Lemma A.1 gives

$$w\left(\frac{z(A_{n,k})}{z(B_k)} - n\right) \geq w(z(B_k)) \geq k + 1.$$

Taking the trace divided by 2 preserves this bound. This proves the first assertion for positive, negative, and zero n. □

Proposition 5.4. *There is a fixed existential formula $\mathrm{Cong}_3(x, T, K)$ such that*

$$\mathrm{Cong}_3(x, nP, kP) \iff v_3(x - n) \geq k \qquad (x \in \mathbb{Q},\ n \in \mathbb{Z},\ k \in \mathbb{N}). \tag{5.7}$$

The formula is false if either point argument is not a valid code for an element of G, or if the index of K is negative.

Proof. The relation

$$\{(k, \nu) \in \mathbb{Z}^2 : k \geq 0,\ \nu = 16 \cdot 3^k\}$$

is computable. Use its fixed definition on G^2 from Corollary 4.10 to obtain $B = B_k$ from $K = kP$, and use $\mathrm{Mul}_G(T, B, A)$ to obtain $A = A_{n,k}$. Introduce c, b by the rational-function identities in (5.5). Finally require the existence of $u \in \mathbb{Z}_{(3)}$ with

$$x - c = bu.$$

Membership in this fixed local ring is existential by Lemma C.1, and $b \neq 0$ by Lemma 5.3. Thus the last condition is equivalent to $v_3(x - c_{n,k}) \geq k$. Since $v_3(c_{n,k} - n) \geq k + 1$, it is equivalent to $v_3(x - n) \geq k$.

Require both free point codes to lie in G. The graph used to choose B has $k \geq 0$ as part of its definition, so the stated domain restrictions follow. Exponentiation appears only in this one fixed relation on integer indices; no variable exponent occurs in a ring term of the formula. □

5.3. **Height control and scalar evaluation.** Write the four finite defining forms from Section 2 as $L_i(U, V) = \alpha_i U + \beta_i V$, with $\alpha_i, \beta_i \in \mathbb{Z}$ and $\alpha_i \neq 0$. For $n \in \mathbb{Z}$ put

$$m_i(n) = \alpha_i n + \beta_i, \qquad R(n) = \prod_{i=1}^{4} m_i(n), \qquad \kappa(n) = H_* R(n)^{2c} + n^2 + 1. \tag{5.8}$$

The product $R(n)$ is signed. The function κ is an integer-coefficient polynomial and is positive on every integer. By Theorem 4.9, its graph on G is defined by the fixed formula $\Theta_{Y-\kappa(X)}(T, K)$:

$$\Theta_{Y-\kappa(X)}(nP, kP) \iff k = \kappa(n).$$

This formula uses only the fixed arithmetic circuit for the displayed polynomial.

Lemma 5.5. *Let $n \in \mathbb{Z}$ with $R(n) \neq 0$. If $x \in \mathbb{Q}$ satisfies*

$$\mathrm{Supp}(\alpha_i x + \beta_i, m_i(n)P) \qquad (1 \le i \le 4), \tag{5.9}$$

then $x \notin \mathcal{C}$ and

$$h(x) \le H_* |R(n)|^c. \tag{5.10}$$

Proof. Each value $\alpha_i x + \beta_i$ is nonzero by the definition of Supp, so the finite point x is not marked. Fix a prime $q \notin S$ not dividing $R(n)$. Proposition 5.2 implies that every valuation $v_q(\alpha_i x + \beta_i)$ is even. Put $\mu_q = \min\{v_q(x), 0\}$. If $v_q(x) \ge 0$, then $\mu_q = 0$. If $v_q(x) < 0$, each α_i is a q-adic unit and each β_i is integral, so

$$v_q(\alpha_i x + \beta_i) = v_q(x).$$

Thus μ_q is even in either case. The four finite normalized contact orders

$$v_q(\alpha_i x + \beta_i) - \mu_q$$

and the contact order $-\mu_q$ at infinity are all even. Consequently every prime in $M_S(x)$ divides $R(n)$, and

$$M_S(x) \mid \prod_{q \mid R(n)} q, \qquad M_S(x) \le |R(n)|.$$

The height estimate (2.6) proves (5.10). □

Let

$$\mathcal{E} = \{n \in \mathbb{Z} : R(n) = 0\} = \mathcal{C} \cap \mathbb{Z}.$$

This is a fixed finite set, possibly empty.

Definition 5.6. For a rational number x and a point code T, require $T \in G$. Define $\mathrm{Eval}(x, T)$ to hold if either

$$\bigvee_{e \in \mathcal{E}} (x = e \ \wedge \ T = eP)$$

holds, or there exist $M_1, \dots, M_4, K \in G$ satisfying

$$\begin{aligned} M_i &= [\alpha_i]T + [\beta_i]P && (1 \le i \le 4), \\ &\mathrm{Supp}(\alpha_i x + \beta_i, M_i) && (1 \le i \le 4), \\ &\Theta_{Y - \kappa(X)}(T, K), \qquad \mathrm{Cong}_3(x, T, K). \end{aligned} \tag{5.11}$$

The empty exceptional disjunction is false.

All multipliers in the first line of (5.11) are fixed integers. The quantifiers range over rational point codes and rational witnesses for the stated formulas.

Theorem 5.7 (Scalar evaluation)**.** *Under Inputs E, H, R, and P, the relation of Definition 5.6 is Diophantine over $\mathbb{Q}$ and satisfies*

$$\mathrm{Eval}(x, nP) \quad \Longleftrightarrow \quad x = n \qquad (x \in \mathbb{Q}, \ n \in \mathbb{Z}).$$

Proof. Let $x = n \in \mathbb{Z}$ and $T = nP$. If $n \in \mathcal{E}$, use the exceptional branch. Otherwise choose $M_i = m_i(n)P$ and $K = \kappa(n)P$. Proposition 5.2 supplies all four support conditions, the polynomial relation gives the required precision point, and Proposition 5.4 applies because $x - n = 0$. Thus every integer has witnesses with its prescribed point nP.

Conversely, fix $T = nP$ and suppose $\mathrm{Eval}(x, T)$ holds. In the exceptional branch, $T = eP$ implies $n = e$ because P is nontorsion, and hence $x = n$. In the other branch the point identities force $M_i = m_i(n)P$. The support conditions make these points nonidentity, so $R(n) \neq 0$. Lemma 5.5 yields $h(x) \le H_* |R(n)|^c$, and the last two conditions give

$$v_3(x - n) \ge \kappa(n).$$

If $x \neq n$, the reduced numerator of $x - n$ is divisible by $3^{\kappa(n)}$. Since $|R(n)| \geq 1$ and $h(n) \leq n^2$, we obtain

$$\begin{aligned}\kappa(n)\log 3 &\leq h(x-n)\\ &\leq h(x) + h(n) + \log 2\\ &\leq H_*R(n)^{2c} + n^2 + \log 2\\ &< H_*R(n)^{2c} + n^2 + 1 = \kappa(n).\end{aligned}$$

This contradicts $\log 3 > 1$ and $\kappa(n) > 0$. Therefore

$$\mathrm{Eval}(x, nP) \quad \Longleftrightarrow \quad x = n.$$

Every relation in Definition 5.6 has a fixed existential definition over $\mathbb{Q}$. In particular, the numbers of point variables, support certificates, and local conditions are independent of n and of the precision $\kappa(n)$. This proves the claimed Diophantine nature of the evaluation graph. □

6. Proof of the equivalence

Proof of Theorem 1.1. Assume E, H, and R with the shared data of Section 2.

From P to $T_{\mathbb{Z}}$. Assume P and choose its single formula Φ and finite set S_Φ. The constructions in Sections 3–5 use these fixed choices and the three arithmetic inputs to prove Theorem 5.7. Hence the positive existential formula

$$\exists T \in G \quad \mathrm{Eval}(x, T)$$

holds exactly when $x \in \mathbb{Z}$. Here T is a fixed tuple of rational coordinates; membership in G has the existential definition in Appendix B. Indeed, an integer n has the witness $T = nP$, and any witness belongs to $G = \mathbb{Z}P$ and therefore has a unique index $n \in \mathbb{Z}$, which the evaluation theorem forces to equal x.

Expand the fixed formulas and point charts, replace inequations by equations with reciprocal witnesses, and put all existential quantifiers in front. Over $\mathbb{Q}$, a conjunction of equations is expressed by a sum of their squares and a disjunction by their product. These operations give one polynomial equation in x and finitely many rational witnesses. Clearing the fixed coefficient denominators gives $D \in \mathbb{Z}[X, Y_1, \ldots, Y_N]$ as in (1.2); an unused witness may be added if necessary to arrange $N \geq 1$. Every formula, coefficient, and witness count is fixed independently of x. This proves $T_{\mathbb{Z}}$.

From $T_{\mathbb{Z}}$ to P. Let D and N satisfy (1.2). Define

$$\Phi(t): \quad \exists \mathbf{y} \in \mathbb{Q}^N \; D(t, \mathbf{y}) = 0, \qquad S_\Phi = \varnothing.$$

This is one positive existential formula over $\mathbb{Q}$. Every integer is accepted. If a rational number is accepted, it is an integer, so its valuation at every rational prime is nonnegative (or is $+\infty$ for zero). The implication in (1.1) therefore holds at every prime. This proves P without using E, H, or R. □

The forward argument uses E to encode and compare signed integer indices, R to supply simultaneous witnesses for integer coordinate representatives, and H to bound heights from the resulting support conditions. The independent reverse argument supplies the existence of some parity predicate; it does not verify any separately proposed candidate formula. The theorem proves the equivalence of the assertions, and no inference here establishes either assertion from E, H, and R alone.

7. Consequences of either equivalent assertion

Throughout this section assume E, H, and R and either P or $T_{\mathbb{Z}}$. By Theorem 1.1, P holds under these assumptions, so the constructions above apply. The integer model transfers integer Diophantine problems to rational ones. Scalar evaluation gives the stronger conclusion that the usual rational parameters may be used directly in such representations.

Corollary 7.1 (A computable many-one reduction)**.** *Under the assumptions of this section, there is a total computable map $f \mapsto Q_f$ from integer-coefficient polynomials to integer-coefficient polynomials such that*

$$\exists \mathbf{n} \in \mathbb{Z}^r \ f(\mathbf{n}) = 0 \quad \Longleftrightarrow \quad \exists \mathbf{y} \in \mathbb{Q}^{N(f)} \ Q_f(\mathbf{y}) = 0.$$

The output may be chosen to have total degree at most four. Consequently $\mathrm{H10}(\mathbb{Q})$ *is* Σ_1^0*-complete under computable many-one reductions.*

Proof. Use the arithmetic-circuit formula of Theorem 4.9 and quantify its input points in G. Expand the fixed predicates and the point codes, replace inequations by reciprocal equations, and combine the resulting finite positive existential formula into one polynomial equation. For a constant input polynomial, use an always true or always false equation according as the constant is zero or nonzero. This is an effective procedure once the fixed data have been chosen.

For the degree bound, take an arithmetic circuit evaluating the resulting polynomial, introduce a variable for each gate output, and impose its addition, constant, or multiplication equation. These equations have degree at most two. Add the equation setting the final output to zero, clear coefficient denominators, and take the sum of their squares. Over $\mathbb{Q}$ the sum vanishes exactly when all the gate equations vanish, and its total degree is at most four.

Here $\mathrm{H10}(R)$ denotes the set of codes of integer-coefficient polynomials having a zero over R, with the number of variables allowed to vary. The Davis–Putnam–Robinson–Matiyasevich theorem makes $\mathrm{H10}(\mathbb{Z})$ complete for computably enumerable sets [2]. Our construction gives a many-one reduction to $\mathrm{H10}(\mathbb{Q})$. The latter set is computably enumerable by enumerating rational tuples and evaluating the polynomial. These two facts give the stated completeness. □

A many-one reduction produces one target instance from each input. No collapse of general Turing reducibility to many-one reducibility is asserted. Here the reduction follows from the explicitly constructed model. Its computability asserts neither a practical running time nor a bound on the size of witnesses. The fixed constants guaranteed by the hypotheses can be included in one algorithm; their numerical values and an executable implementation are not supplied.

Corollary 7.2 (Computably enumerable rational sets)**.** *Under the assumptions of this section, every computably enumerable subset of $\mathbb{Q}^r$ is Diophantine over $\mathbb{Q}$.*

Proof. Use any standard effective encoding of rational numbers by reduced fractions. If $A \subseteq \mathbb{Q}^r$ is computably enumerable, then

$$\{(\mathbf{u}, \mathbf{v}) \in \mathbb{Z}^{2r} : v_i \neq 0 \text{ for all } i, \ (u_1/v_1, \ldots, u_r/v_r) \in A\}$$

is a computably enumerable integer relation. By the Davis–Putnam–Robinson–Matiyasevich theorem it has a Diophantine definition over $\mathbb{Z}$. Theorem 5.7 restricts each of its integer variables to the usual integers inside $\mathbb{Q}$. Adjoin the equations $x_i v_i = u_i$ and the inequations $v_i \neq 0$. This is a Diophantine definition of A over $\mathbb{Q}$. □

Corollary 7.3 (Nonzero witnesses with unchanged parameters)**.** *Under the assumptions of this section, every computably enumerable relation $\mathcal{R} \subseteq \mathbb{N} \times \mathbb{Q}^\times$ has a polynomial*

$$M_{\mathcal{R}}(X, Y, Z_1, \ldots, Z_s) \in \mathbb{Z}[X, Y, Z_1, \ldots, Z_s]$$

such that, for $n \in \mathbb{N}$ and $b \in \mathbb{Q}^\times$,

$$(n, b) \in \mathcal{R} \implies \exists \mathbf{z} \in (\mathbb{Q}^\times)^s \ M_{\mathcal{R}}(n, b, \mathbf{z}) = 0,$$
$$(n, b) \notin \mathcal{R} \implies \nexists \mathbf{z} \in \mathbb{Q}^s \ M_{\mathcal{R}}(n, b, \mathbf{z}) = 0.$$

The parameters are substituted directly as $X = n$ and $Y = b$.

Proof. Declare the relation false outside its specified domain and apply Corollary 7.2. Let $P_{\mathcal{R}}(X, Y, W_1, \ldots, W_e)$ be one defining polynomial with integer coefficients. Replace each W_i by $U_i - V_i$. Every rational number w is the difference of two nonzero rationals: choose $v \in \mathbb{Q} \setminus \{0, -w\}$ and put $u = w + v$. Thus every original witness lifts to a witness with all U_i, V_i

nonzero. Conversely every rational zero after substitution gives an original rational zero, even if some substituted variables are zero. Taking $s = 2e$ proves both assertions; the case $e = 0$ is included. □

The polynomial in the last corollary is attached to one fixed relation $\mathcal{R}$. It is not asserted to be one polynomial simultaneously coding every relation. Any further application, for example to a specified Ramsey-theoretic decision problem, must give the relation and the reduction and retains the same assumptions of this section.

The construction separates arithmetic on point indices, identification of an index with its rational scalar, and compilation into polynomial equations. The first two are proved from E, H, R, and the antecedent P; the last is a finite algebraic operation. The equivalence theorem shows that, under the three stated arithmetic assumptions, the existence of the pole-parity predicate is exactly as strong as $T_{\mathbb{Z}}$. The consequences in this section retain the assumption that one of these equivalent assertions holds.

Appendix A. Formal parameters and the estimates used in the proof

The distinction between two local arguments is useful. For completeness of the support relation, a point need only approach the identity at finitely many places; this works even at bad reduction. For the comparisons used in multiplication and at 3, we use the stronger valuation estimates available at odd unramified places.

Lemma A.1 (Local neighborhoods of the identity)**.** *For the displayed equation of E, at every finite place w there are arbitrarily small open subgroups of $E(F_w)$ parametrized by sufficiently small values of $z = -x/y$. In this parameter the group law has integral coefficients, and for every signed integer a,*

$$[a](Z) = aZ + Z^2 F_a(Z), \qquad F_a(Z) \in \mathcal{O}_{F_w}[[Z]]. \tag{A.1}$$

The function $1/x$ extends to the identity with value zero, and $z([a]T)/z(T)$ extends there with value a. Each of the indicated open subgroups has finite index in $E(F_w)$.

At a place of good reduction, z parametrizes the whole kernel of reduction by the maximal ideal. For a nonidentity point with $x \neq 0$, membership in this kernel is equivalent to $w(1/x) > 0$.

Proof. The Weierstrass coefficients are integral at every finite place. The usual formal construction at the smooth point O therefore gives an integral group law, whether or not the rest of the reduced cubic is smooth [3, Chapters IV and VII]; see also [4, proofs of Lemmas 9 and 10] for the integral parameter expansions and the local subgroups used here. One can see the parameter expansion directly by putting $w_0 = -1/y$:

$$x = z/w_0, \qquad w_0 = z^3 - d^2 z w_0^2.$$

The second equation has a unique solution $w_0 \in z^3\mathcal{O}_{F_w}[[z]]$ with leading term z^3, found recursively. Hence $1/x = w_0/z = z^2 + \cdots$ extends with value zero. The integral addition and inverse series give (A.1), including negative a, and show that sufficiently small parameter balls are subgroups. Their power series converge there. Dividing (A.1) by Z proves the stated extension of the quotient.

These parameter balls are open neighborhoods of O in the analytic group $E(F_w)$. This group is compact because E is projective. An open subgroup of a compact group has finite index: its cosets form an open cover, and a finite subcover contains all cosets. At good reduction the formal group identifies the full kernel with the maximal ideal. The expansions give $w(x) = -2w(z) < 0$ there. Conversely, a pole of x reduces to the unique point at infinity on the smooth Weierstrass model. This proves the last assertion. □

Lemma A.2 (Valuation preservation and the logarithmic unit)**.** *Let q be odd and let w be an unramified place above q, normalized by $w(q) = 1$. Suppose the curve has an integral equation at w. For Z in the maximal ideal, the formal logarithm converges; for $Z \neq 0$,*

$$\ell(Z)/Z \in 1 + q\mathcal{O}_{F_w}, \qquad w(\ell(Z)) = w(Z). \tag{A.2}$$

On the formal group the logarithm is a homomorphism. These assertions hold also for the conjugate curve.

Proof. Write the invariant differential in the formal parameter as $(1+\sum_{j\ge1} b_j Z^j)\,dZ$, with b_j integral. Integrating gives

$$\ell(Z) = Z + \sum_{t\ge2} c_t Z^t, \qquad w(c_t) \ge -v_q(t).$$

The value group is $\mathbb{Z}$. For $w(Z) \ge 1$ and $t \ge 2$,

$$w(c_t Z^{t-1}) \ge (t-1)w(Z) - v_q(t) \ge 1.$$

The bounds tend to infinity; the last inequality follows from $t - 1 - v_q(t) \ge 1$ for odd q. This proves convergence and (A.2). The formal identity defining the logarithm is an identity of convergent series on this group and therefore makes it a homomorphism. Conjugation changes none of the estimates. □

Lemma A.3 (Ratios of parameters and truncated logarithms)**.** *Let q be odd and unramified in F, and suppose the displayed curve has good reduction at $w \mid q$. Normalize $w(q) = 1$. If $U = aP$ and $V = bP$ are formal at w, where $a, b \in \mathbb{Z}$ and $b \ne 0$, then*

$$w\left(\frac{z(U)}{z(V)}\right) = v_q(a/b). \tag{A.3}$$

For every integer $k \ge 1$ with $q > k+1$, if $v_q(a/b) \ge 0$, then

$$\ell_k(z(V)) \ne 0, \qquad w\left(\frac{\ell_k(z(U))}{\ell_k(z(V))} - \frac{a}{b}\right) \ge k. \tag{A.4}$$

The same statements hold on E^σ with P^σ.

Proof. The group identity $[b]U = [a]V$ and Lemma A.2 give $b\ell(z(U)) = a\ell(z(V))$. Since $b \ne 0$ and P is nontorsion, $V \ne O$ and $\ell(z(V)) \ne 0$. Taking valuations proves (A.3), with the infinite-valuation convention when $a = 0$.

For $t > k$ and $q > k+1$,

$$t - 1 - v_q(t) \ge k.$$

Indeed this is immediate when $v_q(t) = 0$; if $j = v_q(t) \ge 1$, then $t-1-j \ge q^j - 1 - j \ge q - 2 \ge k$. The coefficient bound in the preceding proof consequently gives

$$w(\ell(Z) - \ell_k(Z)) \ge w(Z) + k \qquad (w(Z) \ge 1). \tag{A.5}$$

In particular $w(\ell_k(z(V))) = w(z(V))$, proving nonvanishing. Put $\lambda = a/b$. If $a \ne 0$ and $v_q(\lambda) \ge 0$, then $w(z(U)) \ge w(z(V))$ by (A.3). Subtract the two logarithmic tails in $\ell(z(U)) = \lambda\ell(z(V))$ and divide by $\ell_k(z(V))$. Equation (A.5) gives the claimed error at least k. If $a = 0$ the comparison is exact. The conjugate proof is identical. □

The coefficients of ℓ_k are effective elements of F. The recursion for w_0 above determines the invariant differential $dx/(2y)$ to any prescribed order, and termwise integration determines ℓ_k. Thus the fixed polynomial ℓ_c used in the main text requires finitely many rational operations. The integer k in the truncation lemma is positive; the separate congruence precision in Section 5 may equal zero.

Appendix B. Explicit rational encoding of the field and elliptic points

We record finite existential encodings for the algebraic operations used in the proof. Every variable in these encodings ranges over $\mathbb{Q}$; an F-valued variable or a point variable abbreviates a fixed tuple of rational variables.

B.1. **The quadratic field and its rational projection.** Represent $u = u_0 + u_1\sqrt{2} \in F$ by $(u_0, u_1) \in \mathbb{Q}^2$. Then

$$\begin{aligned}(u_0, u_1) + (v_0, v_1) &= (u_0 + v_0, u_1 + v_1),\\ (u_0, u_1)(v_0, v_1) &= (u_0v_0 + 2u_1v_1, u_0v_1 + u_1v_0),\\ (u_0, u_1)^\sigma &= (u_0, -u_1).\end{aligned}$$

Thus an F-equation is a pair of rational equations. The rational projection used in the main argument is

$$\rho(u_0 + u_1\sqrt{2}) = u_0 = \frac{1}{2}\operatorname{Tr}_{F/\mathbb{Q}}(u).$$

The field norm is $\mathrm{N}_{F/\mathbb{Q}}(u) = u_0^2 - 2u_1^2$, which vanishes exactly when $u = 0$. Consequently an F-inequation $u \neq 0$ is expressed by $\mathrm{N}_{F/\mathbb{Q}}(u)s = 1$ for an auxiliary $s \in \mathbb{Q}$. The graph of inversion is given by the two rational equations expressing $uv = 1$. These interpretations do not require a rational square root of 2.

B.2. **Canonical point codes.** A code for a point of E consists of a rational flag ϵ and two F-coordinates x, y, hence five rational components. Its validity is the finite condition

$$\mathrm{Pt}_E(\epsilon, x, y): \quad (\epsilon = 0 \ \wedge \ x = y = 0) \ \vee \ (\epsilon = 1 \ \wedge \ y^2 = x^3 - d^2x).$$

The first branch is the unique code for O; the second is the unique affine code for the finite point with coordinates (x, y). In particular, the finite two-torsion point $(0, 0)$ has flag 1 and is distinct from the identity code.

Equality of valid codes is equality of their five rational components. For codes with components $q_1, \dots, q_5$ and $q_1', \dots, q_5'$, inequality is equivalent to

$$\sum_{i=1}^{5}(q_i - q_i')^2 \neq 0.$$

A reciprocal witness makes this an existential polynomial condition. Conjugation acts componentwise by the displayed rational linear map and gives the corresponding code on E^σ, whose equation has d replaced by d^σ.

B.3. **Addition, inverses, and fixed integer multiples.** For valid codes U, V, W, the equation $W = U + V$ is expressed by a finite disjunction of the following cases. If $U = O$, impose $W = V$; if $V = O$, impose $W = U$. In every remaining branch require U and V to be finite. If $x_U = x_V$ and $y_U + y_V = 0$, impose $W = O$; this includes doubling a two-torsion point.

If $x_U \neq x_V$, require W to be finite and introduce $\lambda \in F$ satisfying

$$\begin{aligned}(x_V - x_U)\lambda &= y_V - y_U,\\ x_W &= \lambda^2 - x_U - x_V,\\ y_W &= \lambda(x_U - x_W) - y_U.\end{aligned}$$

If $x_U = x_V$, $y_U = y_V$, and $y_U \neq 0$, require W to be finite and use

$$\begin{aligned}2y_U\lambda &= 3x_U^2 - d^2,\\ x_W &= \lambda^2 - 2x_U,\\ y_W &= \lambda(x_U - x_W) - y_U.\end{aligned}$$

These cases exhaust the group law on the nonsingular curve: finite points with the same x-coordinate have equal or opposite y-coordinates. All the displayed equations and inequations have the rational encodings already described. These are the usual addition formulas [3, Chapter III, §2].

Inversion fixes O and sends (x, y) to $(x, -y)$ on the finite chart. For any fixed integer a, a fixed finite sequence of additions and, if needed, an inversion defines the graph of $[a]$. Thus membership in the chosen group $G = m_0E(F)$ is expressed by

$$\mathrm{Pt}_E(T) \ \wedge \ \exists U\big(\mathrm{Pt}_E(U) \ \wedge \ [m_0]U = T\big),$$

where m_0 is the fixed integer from Input E. No negation of this existential membership condition is used.

B.4. **Rational functions and their domains.** On the identity branch define $z(O) = 0$. On a finite branch with $y \neq 0$, the equation $yz = -x$ defines $z = -x/y$. On a finite branch with $x \neq 0$, the equation $xt = 1$ defines $t = 1/x$. A quotient A/B is represented by a variable D and the conditions $B \neq 0$, $BD = A$, over F or $\mathbb{Q}$ as appropriate. The projection ρ then selects the first rational component of its F-argument.

Equations for a rational function remain inside the same disjunct as their domain restrictions. In particular, reciprocal-coordinate equations are not imposed on identity branches. All points to which $1/x$ or a nonzero z-denominator is applied in the main argument belong to $G \setminus \{O\}$ or $G^\sigma \setminus \{O\}$, on the corresponding curve. Their coordinates x, y are nonzero, since points with $x = 0$ or $y = 0$ on either displayed curve are two-torsion and both groups are torsion-free. The witnesses used to encode field arithmetic, point operations, and inverses are not subjected to pole-parity conditions unless a predicate explicitly lists them. Each abbreviation therefore expands into a finite existential combination of rational polynomial equations and inequations.

Appendix C. An explicit definition of the fixed local ring

The congruence construction needs only one fixed local ring. The following formula is a direct specialization of classical local–global theory for quadratic forms. We give its proof, including the local obstruction at 3.

Lemma C.1. *For every $t \in \mathbb{Q}$,*

$$t \in \mathbb{Z}_{(3)} \quad \Longleftrightarrow \quad \exists u, v, w \in \mathbb{Q} : \quad u^2 + v^2 + 3w^2 = 1 + 6t^2,$$

where $\mathbb{Z}_{(3)} = \{t \in \mathbb{Q} : v_3(t) \geq 0\}$.

Proof. Put $q(U, V, W) = U^2 + V^2 + 3W^2$. First consider its values over $\mathbb{Q}_3$. If u, v are not both zero and $a = \min\{v_3(u), v_3(v)\}$, then

$$v_3(u^2 + v^2) = 2a.$$

Indeed, after division by 3^{2a}, the two squares reduce to 0 or 1 modulo 3, and at least one reduces to 1; their sum cannot vanish. The valuation of $3w^2$, when finite, is odd. Thus the two summands $u^2 + v^2$ and $3w^2$ have distinct valuations whenever both are nonzero. It follows that whenever $q(u, v, w)$ has odd valuation e, its normalized unit $3^{-e}q(u, v, w)$ is congruent to 1 modulo 3. The same conclusion holds when $u = v = 0$.

If $v_3(t) = -h < 0$, write $t = 3^{-h}\eta$ with $\eta \in \mathbb{Z}_3^\times$ and $h \geq 1$. Then

$$1 + 6t^2 = 3^{1-2h}(3^{2h-1} + 2\eta^2).$$

Its valuation is the odd integer $1 - 2h$, and its normalized unit is 2 modulo 3. It cannot be represented by q over $\mathbb{Q}_3$, and hence cannot be represented over $\mathbb{Q}$. Thus rational witnesses imply $t \in \mathbb{Z}_{(3)}$.

Conversely, suppose $v_3(t) \geq 0$, and put $A = 1 + 6t^2 > 0$. We check that q represents A over every completion of $\mathbb{Q}$. At 3, the number A is congruent to 1 modulo 3, so Hensel's lemma [8, Theorem 2.1] gives $s \in \mathbb{Z}_3$ with $s^2 = A$; take $(u, v, w) = (s, 0, 0)$.

For an odd prime $p \neq 3$, the two subsets

$$\{a^2 : a \in \mathbb{F}_p\}, \qquad \{-3 - b^2 : b \in \mathbb{F}_p\}$$

each have $(p + 1)/2$ elements, so they intersect. We obtain a nonzero solution of $q(U, V, W) = 0$ modulo p with $W = 1$. At least one of U, V is nonzero, because $p \neq 3$; fixing the other two coordinates and applying Hensel's lemma lifts this solution to $\mathbb{Q}_p$. Thus q is isotropic over $\mathbb{Q}_p$. At 2, the unit $-7 \equiv 1 \pmod 8$ is a square in $\mathbb{Q}_2$, so $(\sqrt{-7}, 2, 1)$ is a nonzero isotropic vector [8, Theorem 4.5].

A nondegenerate isotropic quadratic form represents every scalar: if $q(\mathbf{v}) = 0$ with $\mathbf{v} \neq 0$, nondegeneracy supplies $\mathbf{w}$ with $B(\mathbf{v}, \mathbf{w}) \neq 0$, where B is its associated symmetric bilinear form. The expression $q(\mathbf{w} + \lambda\mathbf{v}) = q(\mathbf{w}) + 2\lambda B(\mathbf{v}, \mathbf{w})$ then takes every value as λ varies. Applying

this observation to q proves local representability of A at all finite primes other than 3. Over $\mathbb{R}$, representability follows from $A > 0$.

The homogeneous quadratic form

$$U^2 + V^2 + 3W^2 - AS^2$$

therefore has a nonzero zero over every completion of $\mathbb{Q}$. By the Hasse–Minkowski theorem it has a nonzero zero over $\mathbb{Q}$ [7, Theorem 11.12]. At such a rational zero, S cannot vanish: the form $U^2 + V^2 + 3W^2$ is positive definite over $\mathbb{R}$. Dividing by S^2 gives the required rational witnesses. □

References

[1] OpenAI. *Hilbert's tenth problem over the rational numbers*. Manuscript dated 24 September 2026, repository snapshot adc7f1241b42e322a6451854ab7e4b4c146bf78a. https://github.com/openai/math/tree/adc7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/Hilberts-tenth-problem-over-the-rational-numbers-September-24-2026. The formulations used here correspond to the labels `par:pole-parity`, `ell:rank`, `hei:height`, and `pri:pattern`. The present paper uses E, H, and R as explicit assumptions rather than invoking proofs of them from this source.

[2] Martin Davis. Hilbert's tenth problem is unsolvable. *The American Mathematical Monthly* **80**, no. 3 (1973), 233–269. https://math.umd.edu/~mcl/Pubs/713/Diophantine.pdf.

[3] Joseph H. Silverman. *The Arithmetic of Elliptic Curves*. Second edition, Graduate Texts in Mathematics, vol. 106. Springer, New York, 2009. https://doi.org/10.1007/978-0-387-09494-6.

[4] Bjorn Poonen. Using elliptic curves of rank one towards the undecidability of Hilbert's Tenth Problem over rings of algebraic integers. In C. Fieker and D. Kohel, eds., *Algorithmic Number Theory*, ANTS-V, Lecture Notes in Computer Science, vol. 2369. Springer, Berlin, 2002, pp. 33–42. https://math.mit.edu/~poonen/papers/ants5.pdf.

[5] Bjorn Poonen. *Hilbert's Tenth Problem over Rings of Number-Theoretic Interest*. Lecture notes for the Arizona Winter School, "Number theory and logic," Tucson, 15–19 March 2003. Notes dated 28 February 2003. https://math.mit.edu/~poonen/papers/aws2003.pdf.

[6] Bjorn Poonen and Alexandra Shlapentokh. Diophantine definability of infinite discrete nonarchimedean sets and Diophantine models over large subrings of number fields. *Journal für die reine und angewandte Mathematik* **588** (2005), 27–47. https://math.mit.edu/~poonen/papers/bigsubrings.pdf. Bibliographic record: https://arxiv.org/abs/math/0408271.

[7] Andrew V. Sutherland. *18.782 Introduction to Arithmetic Geometry*, Fall 2013, Lecture 11, 10 October 2013. Massachusetts Institute of Technology. https://math.mit.edu/classes/18.782/2013fa/LectureNotes11.pdf.

[8] Keith Conrad. *Hensel's Lemma*. Expository notes. https://kconrad.math.uconn.edu/blurbs/gradnumthy/hensel.pdf.